

Aegis-Class Rover

Lunar Surface Mobility Platform

Architecture Reference Document | May 2026 | Not for Hardware Commitment

1. Executive Summary

The Aegis-Class Rover is a pressurized, long-duration lunar surface mobility platform designed to support sustained operations at the lunar south pole and other high-value exploration sites. It combines extended-duration crew habitation (30–60 days), integrated power and thermal architecture optimized for low solar elevation, suitport-based dust boundary control, radiation storm shelter capability, and a NASA cFS-based flight software and crew operations console.

ASI develops the vehicle-defining systems: the pressurized crew module, the integrated power and thermal architecture (SSSRA), and the flight software and operations console. The mobility chassis is sourced rather than rebuilt — drawn from the maturing open-architecture lunar rover market — with ASI specifying the interface and integrating the platform.

The vehicle is architected as a surface infrastructure element, not a short-sortie exploration rover. It is intended to operate as part of a distributed logistics network including orbital habitats, surface outposts, LUNET utility nodes, and ISRU systems.

2. Mission Architecture & Operational Concept

Primary Mission Roles

- Polar water prospecting support
- Surface logistics transport
- Mobile command for construction and field operations
- ISRU equipment towing and servicing
- Emergency shelter and crew lifeboat

Operational Regime

Parameter	Value
Nominal crew	2–4
Emergency crew	Up to 6 (short-duration)
Mission duration	30–60 days
Range	~100 km nominal; extended via LUNET recharge (energetic, not geometric)
Cruise speed	10–15 km/h
Maximum speed	25 km/h (terrain limited)

3. Design Philosophy — Constraint Hierarchy

The rover geometry is derived from a strict constraint chain, ensuring traceability from crew habitability to vehicle mass and stability:

- Interior standing height (2.1 m clear)
- Exterior hull dimensions
- Center of gravity (CG) height
- Track width and static stability requirement
- Roof footprint for power/thermal systems
- Radiator and solar array sizing
- Electrical and thermal system capacity

4. Dimensional & Stability Parameters

Exterior Geometry

Parameter	Value
Overall length	~10.0 m
Overall width (with treads)	4.6–4.8 m
Track width (center-to-center)	4.0 m
Wheelbase	6.5 m
Height to cabin roof	3.45 m
Height to array top	3.90 m
Ground clearance	~0.50 m

Stability Metrics

Parameter	Value
CG height	~1.54 m above ground
Static Stability Factor (SSF)	1.30
NASA target (crewed)	≥ 1.2

The 4.0 m track width provides margin for dynamic loads, slope traversal, and cargo asymmetry.

5. Mass Budget (Baseline Configuration)

Subsystem	Mass (kg)
Chassis, suspension, drivetrain	1,200
Pressure hull & structure	1,000
Underfloor systems (batteries, tanks)	800
ECLSS + interior equipment	600
Radiator assembly (6 m²)	60
Roof solar array (20 m²)	110
Deployable wing arrays (10 m²)	75
Fuel cell + reactants	180
Crew (3 × 80 kg)	240
Remaining systems + margin	~1,135
Gross vehicle mass	~5,400

Chassis, suspension, and drivetrain mass is the platform partner's allocation against the integrated-vehicle budget.

6. Mobility Platform Interface

The mobility chassis — suspension, drivetrain, wheels, and body structure — is sourced from a third-party platform partner drawn from the maturing open-architecture lunar rover market. ASI specifies the interface and integrates the platform; the reference configuration below states the operational envelope and interface requirements the crew module brings to any candidate platform. Specific chassis selection and final parameter values follow once a platform partnership is established around this envelope.

Reference Structural Frame

- Rigid metal or composite chassis (partner-defined)
- Multi-axle geometry (6-wheel reference configuration)
- Independent terrain-following suspension
- Wide stance for lateral slope stability

Reference Wheel System

- Non-pneumatic metal-mesh flex wheels
- Designed for regolith abrasion and thermal cycling
- Ground-pressure target ≤ 7 kPa for south-pole regolith

7. Flight Software & Crew Operations Console

The rover flight software is built on NASA's Core Flight System (cFS) — the same flight-heritage framework used across NASA missions. Mission-application source is identical across simulation, bench, and flight targets; only the hardware abstraction layer (HAL) changes between environments, so verification performed in simulation carries forward rather than being rebuilt for each phase.

Development status: The flight software runs end-to-end in a desktop simulation — a real cFS workspace, all 16 applications, with autonomous FDIR verified live across the full safing chain. *Bench and on-vehicle targets share the same source through the HAL; no flight hardware is committed.*

Parameter	Value
cFS applications	16 (cFE core + 9 standard cFS + 7 Aegis mission apps)
FDIR safing modes	5 — NOMINAL → THERM_SAFE → ECLSS_SAFE → LOADSHED → SHELTER
Navigation update rate	20 Hz (aegis_nav navigation & odometry)
Telemetry protocol	CCSDS (standards-compliant)
Console link	WebSocket bridge (CCSDS-UDP ↔ browser)
Runtime	cFE — Core Flight Executive, NASA flight heritage

Autonomous Fault Detection, Isolation & Recovery

- Continuous watchpoint monitoring across thermal, life support, electrical, and radiation domains
- Autonomous transition through the safing chain without requiring crew action
- Each mode a documented operational posture (entry conditions, reconfiguration, crew advisories)
- Crew override authority preserved at all levels

Autonomous Waypoint Navigation

- Waypoint catalog and guidance state machine for autonomous traverse
- Reachability gated by live state-of-charge, vehicle load, and multi-hop LUNET reachability — energetic, not geometric range
- Planner accounts for available recharge stops along candidate routes
- LIDAR + visual SLAM terrain classification; manual, semi-autonomous, and fully autonomous modes; return-to-base; real-time hazard detection

Crew Operations Console

- Browser-based real-time interface for situational awareness and command authority

- Pages: summary status, navigation / waypoints, power / thermal, atmosphere / ECLSS, pilot view, contingency recovery
- Identical across simulation, bench, and flight — consistent operator training and procedure validation

The same flight software and console are designed to operate across whichever mobility chassis the crew module is paired with; the integration spine is consistent across platform partnerships.

8. Command Module & Habitability

Interior Dimensions

Parameter	Value
Interior length	8.0 m
Interior width	3.4 m
Interior height	2.1 m clear
Gross volume	~57 m³
Net habitable volume	~38–42 m³

Functional Zones

- **Zone A:** Forward Command
- **Zone B:** Central Workspace / Galley
- **Zone C:** Crew Quarters / Medical
- **Zone D:** Systems Bay
- **Zone E:** Suitport Vestibule

Cabin Atmosphere

Parameter	Value
Total pressure	55.2 kPa (8.0 psia)
O ₂ partial pressure	21.4 kPa
O ₂ fraction	38.7%
Temperature	18–22°C
Relative humidity	40–55%

Reduced pressure lowers structural mass and reduces EVA prebreathe requirements.

9. Life Support & Consumables

ECLSS Components

- Dual-bed CO₂ scrubber
- Trace contaminant control
- Humidity condensation loop
- Water recovery (85% nominal)
- High-pressure O₂/N₂ storage

Water Budget (3 Crew, 60 Days)

Category	Gross Demand (kg)
Drinking	360
Food rehydration	90
Hygiene	90
Medical / contingency	36
Total gross demand	576

With 85% recovery, net makeup requirement ≈ 86 kg. Vehicle carries ~200 kg at mission start.

10. Suitport & Dust Management

EVA Architecture

- External suitport docking
- No full-cabin depressurization required
- Faster ingress/egress
- Reduced atmosphere loss

Vestibule Systems

- Dedicated HEPA filtration
- Electrostatic dust precipitation
- Sticky mat threshold control
- Cyclonic vacuum
- Mandatory wipe-down protocol

Fallback capability: traditional airlock mode.

11. Radiation Protection

Solar Particle Event Shelter

- Deployable water-wall shielding
- 8–12 cm water equivalent
- Underfloor water tanks provide bottom shielding

Estimated severe SPE dose reduced from lethal exposure range to survivable emergency exposure range.

Galactic Cosmic Radiation

- 0.3–0.6 mSv/day baseline
- 60-day exposure within occupational limits

12. Power Architecture

Electrical Loads

Mode	Power
Cruise	~4,590 W
Station-keeping	~2,170 W
Peak	~8,890 W

Generation Strategy

Source	Capability
Roof SSSRA array (20 m²)	~724 W @ 6° sun
Deployable wing arrays (10 m² tilted)	~3.5 kW combined
Fuel cell	1–3 kW sustained
Battery storage	40–60 kWh
Outpost recharge	LUNET node, logistics-based

13. Thermal Architecture — SSSRA

Radiator Tier

Parameter	Value
Area	6.0 m ²
Operating temperature	310 K
Degraded net rejection	~255 W/m ²
Peak rejection capacity	~1,530 W

Solar Tier

- Roof shading array: 20 m²
- Deployable wing arrays: 10 m²
- Low-emissivity backside coating
- Heading-independent rejection performance

Mass & CG Impact

- Total assembly mass: ~85–110 kg
 - CG rise: ~0.02 m (negligible)
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14. Development Phasing

Current Status

- cFS flight software and crew console running end-to-end in full software simulation
- Autonomous FDIR verified live across the safing chain
- Mobility platform partner selection in progress

Phase 1

- Uncrewed demonstrator
- SSSRA thermal validation
- Dust and suitport testing
- Flight software to bench and hardware-in-the-loop targets through the HAL

Phase 2

- Crewed 7-day sortie capability
- Partial closed-loop ECLSS

Phase 3

- Full 30–60 day mission capability
 - ISRU integration and surface logistics chain
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15. Programmatic Position

The Aegis-Class Rover is designed to:

- Bridge short-duration sorties and permanent surface habitats
 - Enable distributed surface industry
 - Operate as part of a broader cislunar logistics architecture
 - Provide safe, stable, long-duration surface mobility without requiring immediate fixed megastructure deployment
 - Concentrate ASI development on the crew module, integrated power/thermal, and flight software — the vehicle-defining systems — while sourcing the mobility chassis from the open-architecture lunar rover market
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